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Supplementary Information for “Standard compliance bounds origin–destination identifiability in GBFS bike-sharing feeds”

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Abstract

This supplementary information accompanies the manuscript *Standard compliance bounds origin–destination identifiability in GBFS bike-sharing feeds* (main text, Sec. 1–8). Appendix A proves the cost identifiability theorem and the selection-bias decomposition; Appendix B proves the forward-stability lemma and the sampling-horizon law; Appendix C details the constant-estimation procedures used in the pilot.

A Identifiability and the selection-bias decomposition

[To confirm: Proof of main-text Theorem (cost identifiable modulo $\mathcal{N}$) via the two-way fixed-effect / ground-metric argument; proof of $\Pi_{\mathcal{N}^\perp} \hat{c} \rightarrow \Pi_{\mathcal{N}^\perp} c^* - \varepsilon \Pi_{\mathcal{N}^\perp} \log S$, and the corollary that separable selection $S_{ij} = a_i b_j$ gives zero bias while a duration-dependent capture probability $S_{ij} = h(t_{ij})$ does not. Worked example: attenuation of the distance coefficient as a function of Δ .]

B Forward stability and the sampling-horizon law

[To confirm: Sensitivity lemma for the entropic-OT coupling w.r.t. the cost ($\partial P / \partial c$, Hilbert-metric contraction, $O(1/\varepsilon)$); the ε bias–variance balance giving the $n^{-1/4}$ OD rate; and the resulting horizon $T^* \propto \delta_{\text{OD}}^{-4} q^{-1}$ with the standard-compliance threshold. Discuss why the constants degrade at large N and the role of $\lambda_{\min}$.]

C Constant-estimation procedures (pilot)

[To confirm: Estimators for q (identifier persistence), $\kappa(\Delta)$ (aliasing curve by interval subsampling), β (rebalancing fraction), γ (jitter/collision loss), Λ (trips/day), and $\lambda_{\min}$ (projected feature Gram). Cross-reference experiments/d01_pilot.py.]

References

References